

HYPER Obamify

Group 8

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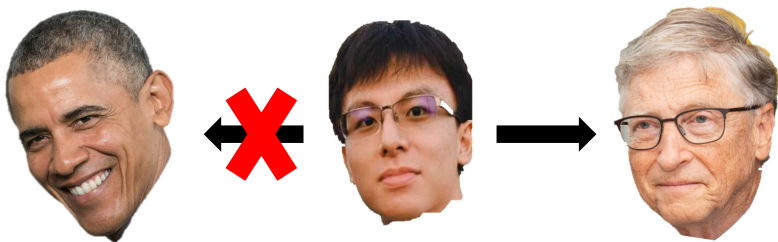
b11901054 吳柏均

Topic

Obamify is an algorithm devised by the YouTuber Spu7Nix.

It turns every random image into Obama.

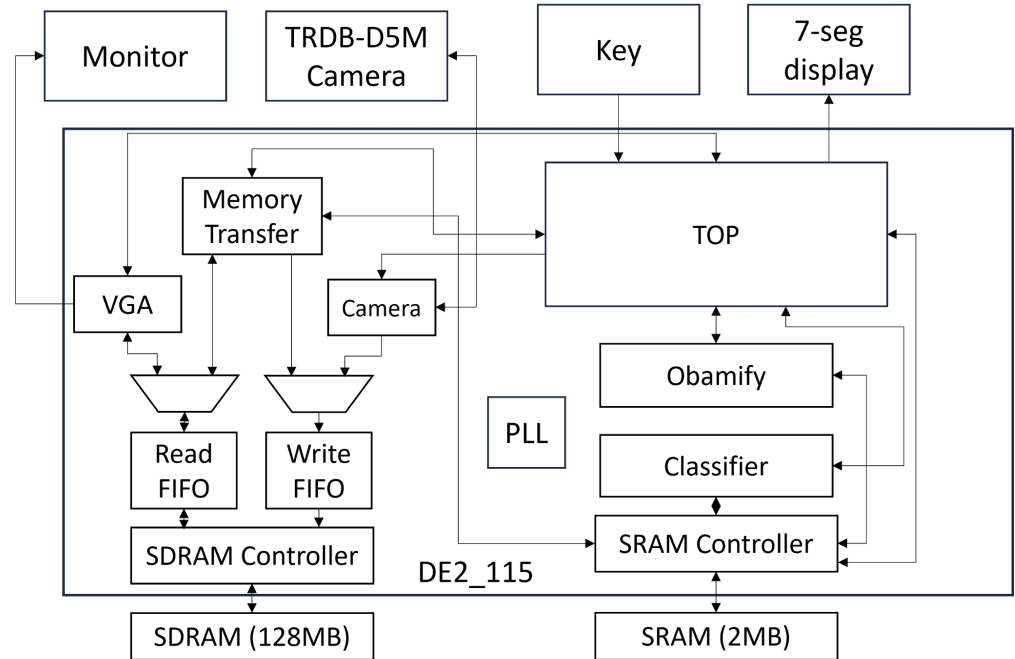
<https://github.com/Spu7Nix/obamify>





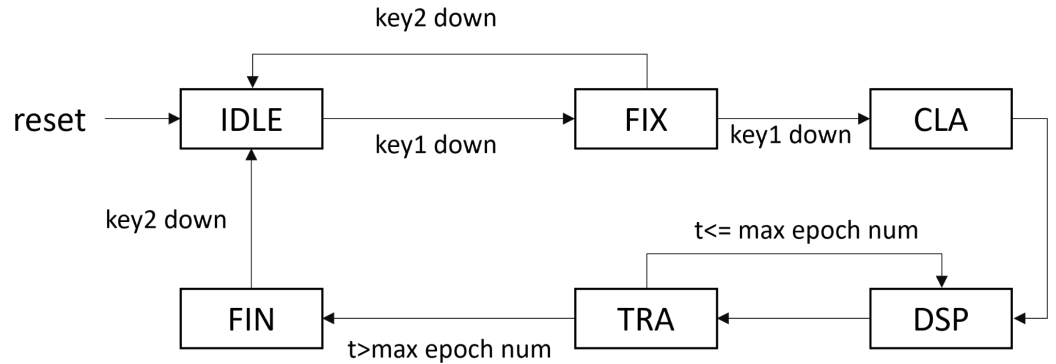
Implementation: Hardware Structure

- Off-board peripheral:
 - monitor
 - camera
- On-board peripheral:
 - key
 - 7 seg display
 - SDRAM
 - SRAM
- Use PLL to provide clock for vga, SRAM, SDRAM, etc.
- Top: state machine and distribute control signal



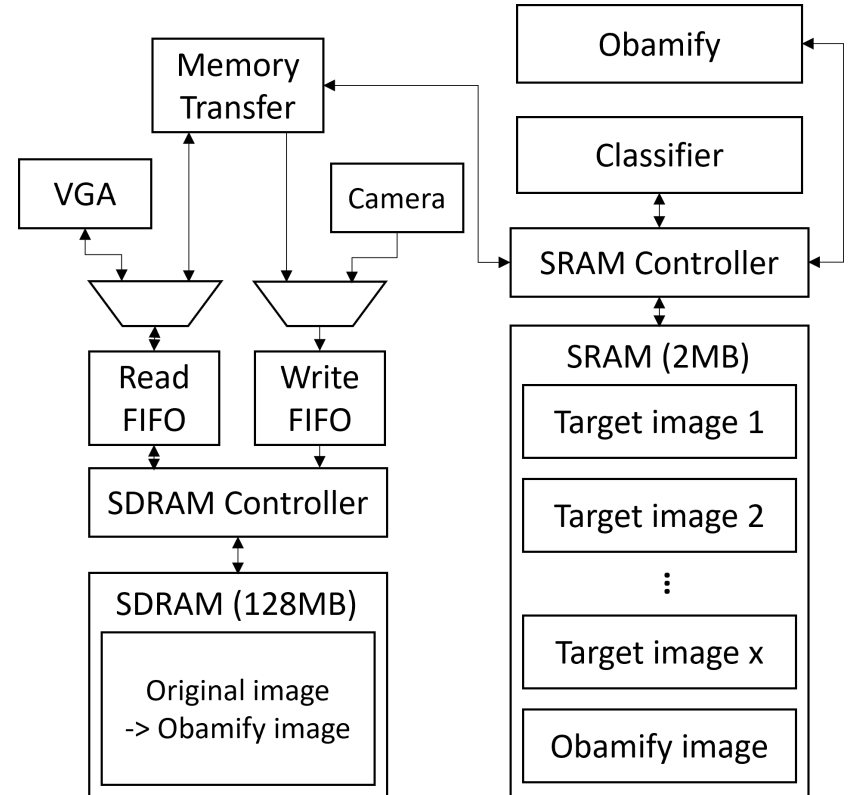
Implementation: State

- **IDLE**: ready to take photo
- **FIX**: fixed image on screen
- **CLA**: classification
- **DSP**: obamify
- **TRA**: transfer file from SRAM to SDRAM
- **FIN**: fixed post-obamify image



Implementation: Memory

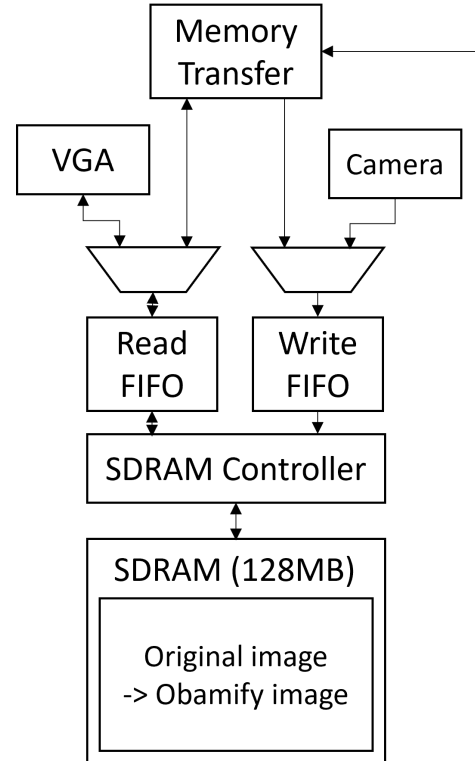
- Use SRAM and SDRAM meanwhile
- SDRAM for external hardware
- SRAM for internal processing



Implementation: Memory

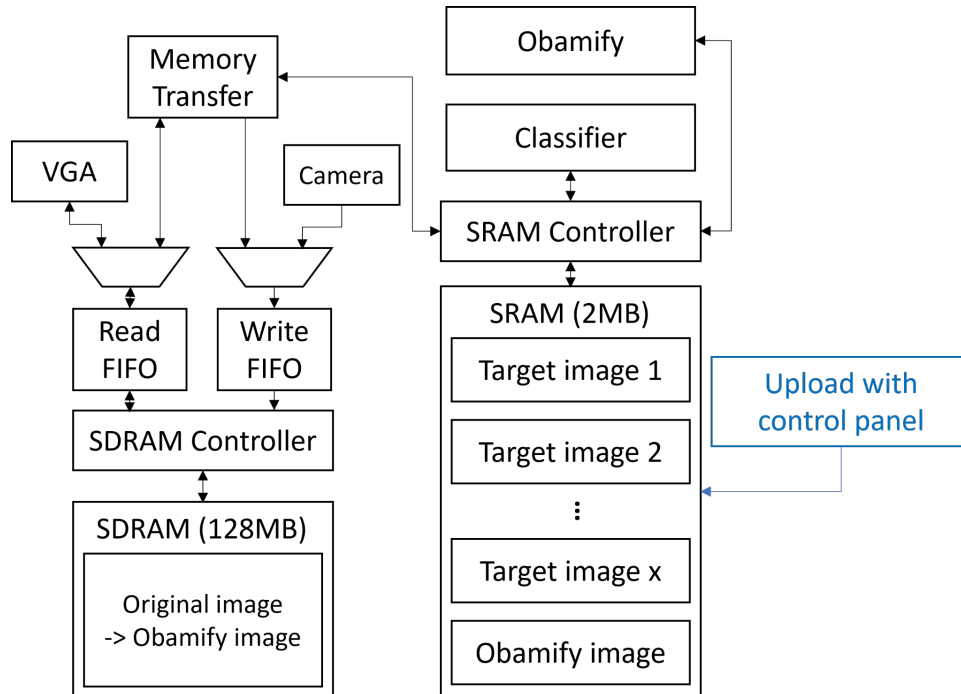
Camera write & VGA read can happen in the same cycle

- **Write FIFO:** When stored word ≥ 128 :
issue write request
- **Read FIFO:** When stored word ≤ 128 :
issue read request
- **SDRAM Controller:** arbit read/write
request, only one access SDRAM at a
time

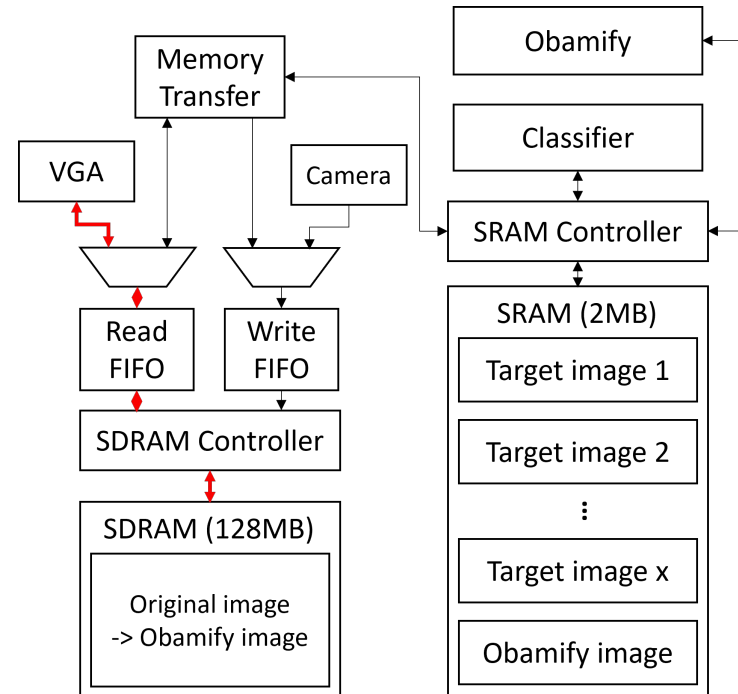


Implementation: Memory

initialize

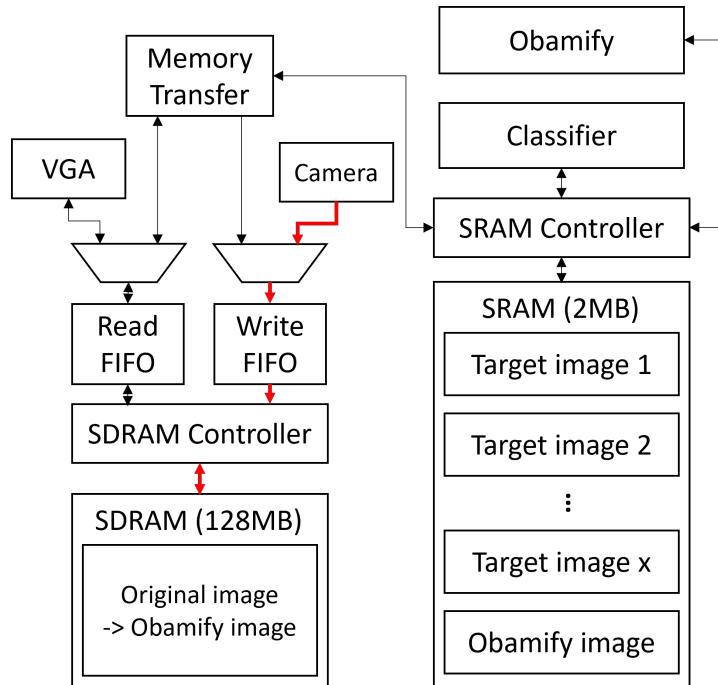


VGA (all state)

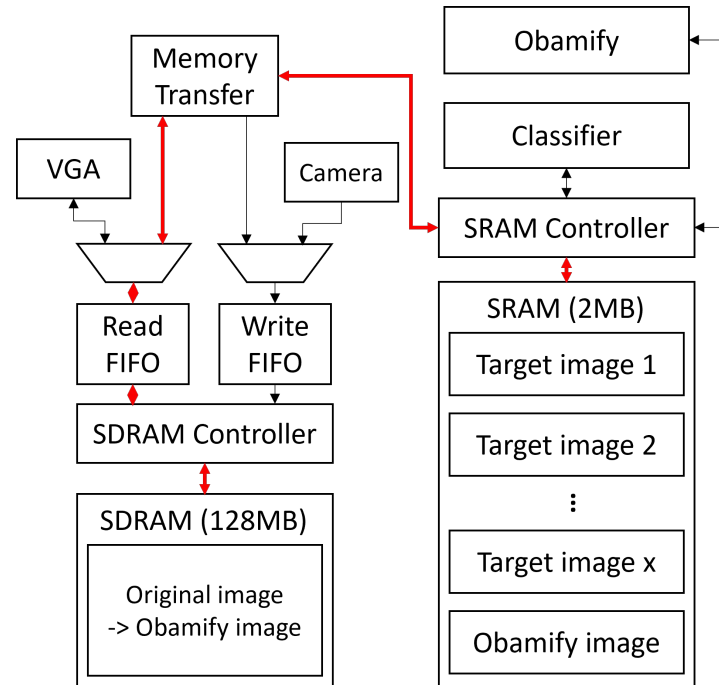


Implementation: Memory

IDLE (camera on)

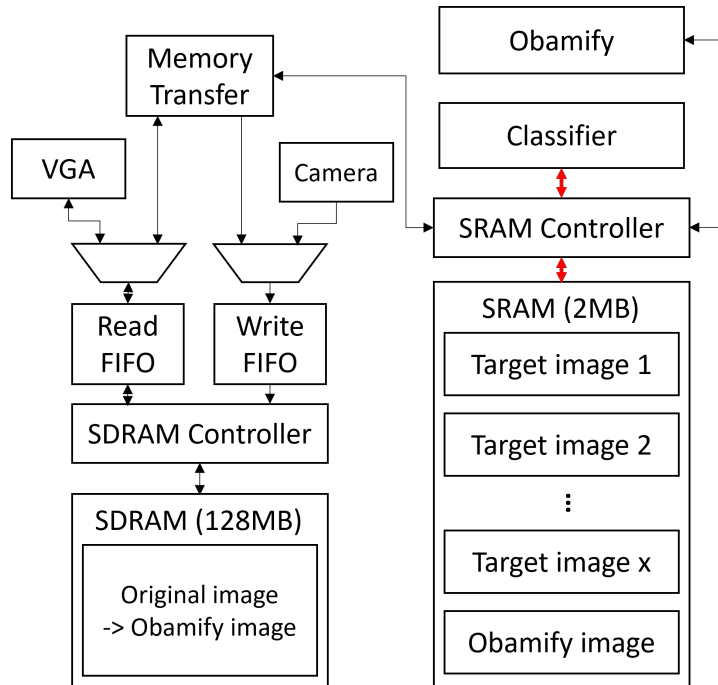


FIX (SDRAM to SRAM)

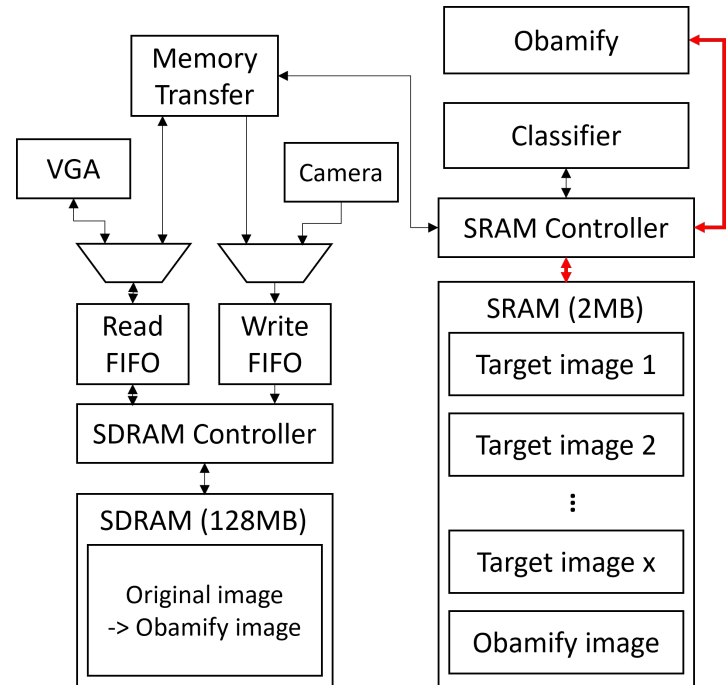


Implementation: Memory

CLA

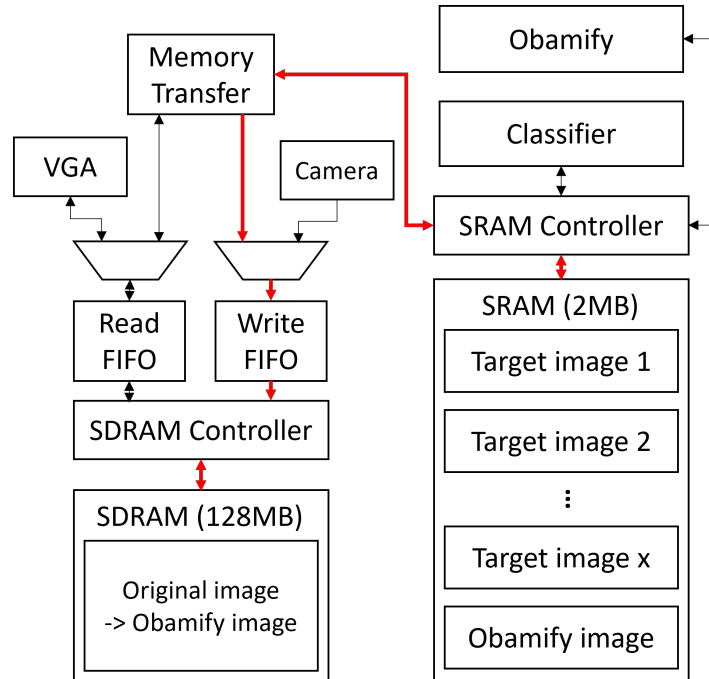


DSP



Implementation: Memory

TRA (SRAM to SDRAM)



Implementation: Camera

Bayer format

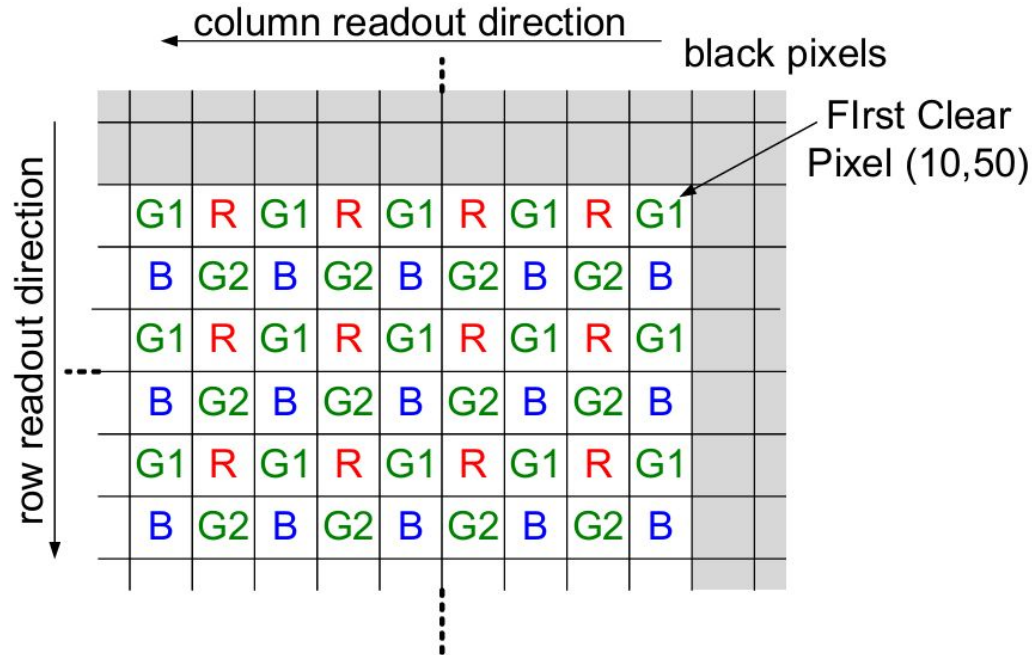


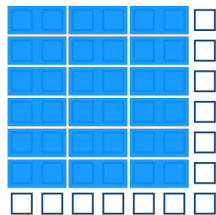
Figure 1.2 Pixel Color Pattern Detail (Top Right Corner)

Implementation: Camera

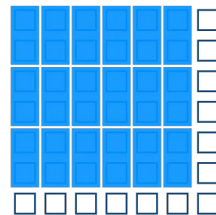
Configurations:

- No zoom mode:
 - Select center 1024x1024 window
 - Binning = 3
- Zoom mode:
 - Select center 512x512 window
 - Binning = 1

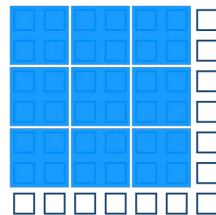
Horizontal Binning



Vertical Binning



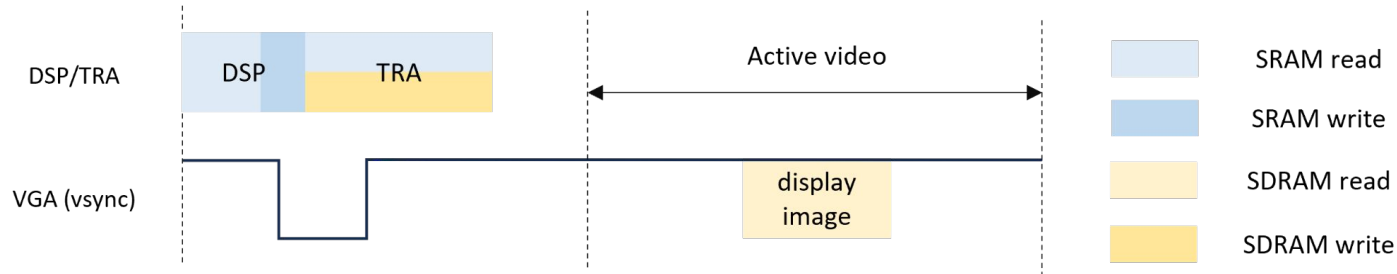
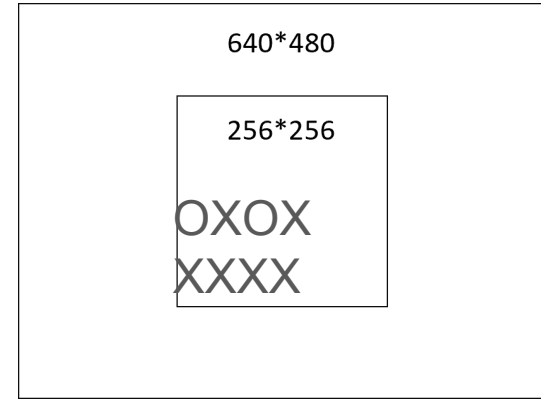
Full Binning



→ Capture and save a 128x128 RGB image

Implementation: VGA

- 60Hz, 640*480 frame
- Run Obamify (DSP) and transfer image to SDRAM in front porch, back porch and sync pulse
- Zoom 128*128 photo to 256*256 with one request in four pixels



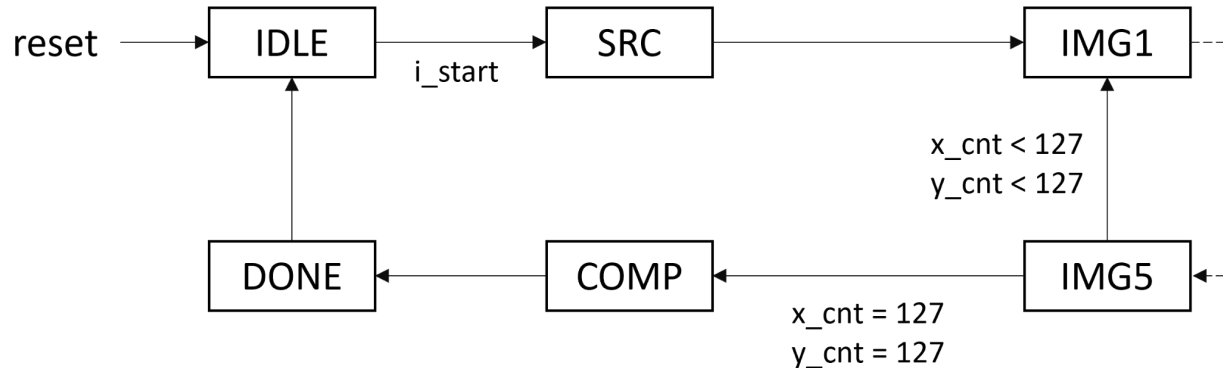
Implementation: Reset Delay

Issue: Memory FIFO needs a short period of time after reset, before other modules can start reading/writing

- RstDelay0: Memory, DSP, Control
- RstDelay1: Camera RGB processing
- RstDelay2: Camera I2C, Camera capture, VGA
- RstDelay3: Camera start

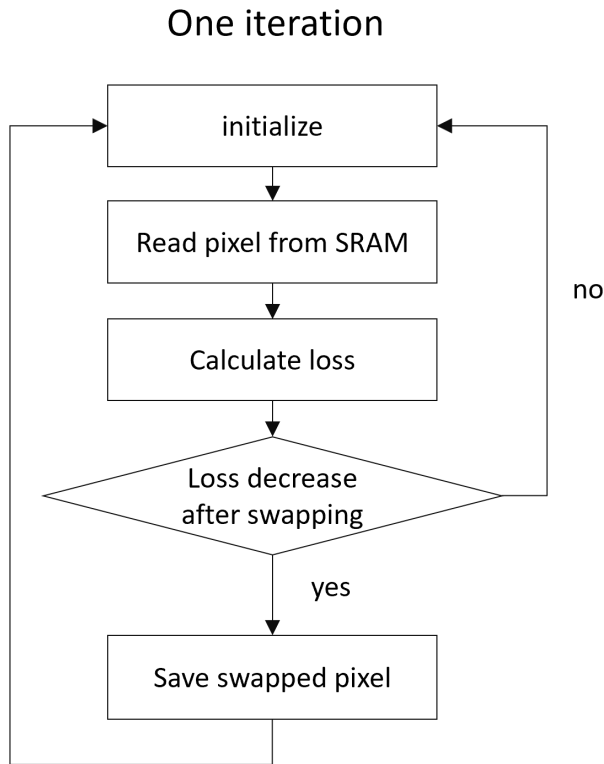
Algorithm: Classify

- **IDLE**: waiting for start
- **SRC**: fetch source image pixel
- **IMG1-5**: fetch source image pixel / calculate SAD (Sum of Absolute Differences)
- **COMP**: compare loss
- **DONE**: output the index of image with minimum index



Algorithm: Obamify

- Calculate loss and change the original image if necessary in an iteration
- whole process: M iterations * N epochs
- use a random generator to decide pixel positions
- reading requires 16 cycles, writing requires 4 cycles
- Loss: $(\Delta R^2 + \Delta G^2 + \Delta B^2) / 256$



Obamify: Loss

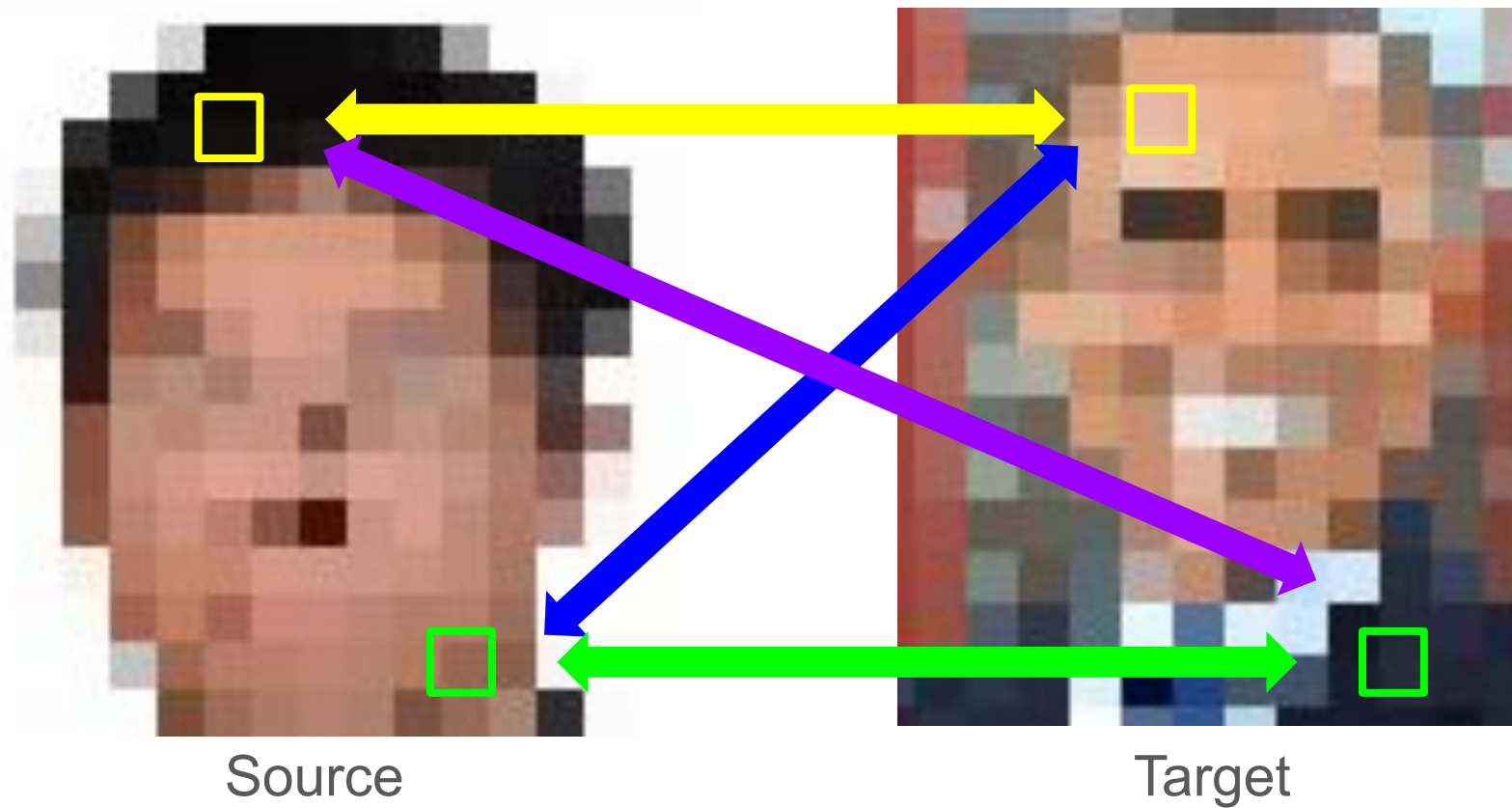


Source

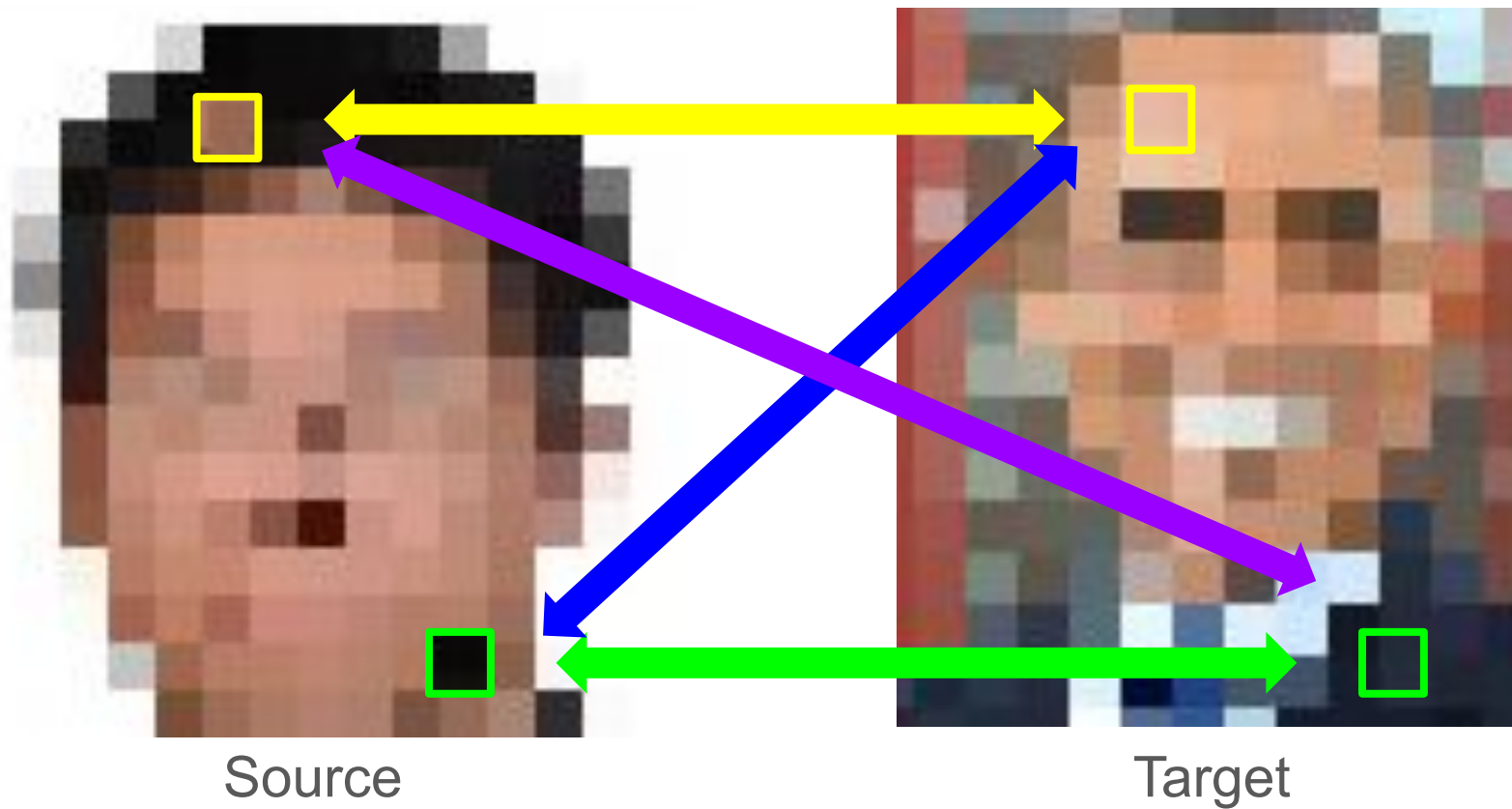


Target

Obamify: Loss



Obamify: Loss



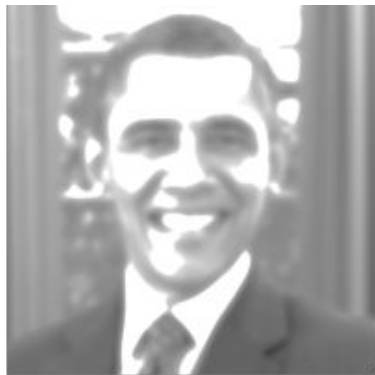
Obamify: Target Image Processing



Original



Head



BG-Blur



BG-Mask



Target

Obamify: Face Swap



Summary

- Use SRAM and SDRAM to decrease simultaneous reading and writing
- Stagger the time for DSP and VGA display
- Multiple candidates and retaking photo function
- Guaranteeing convergence though pixel is picked randomly

DEMO